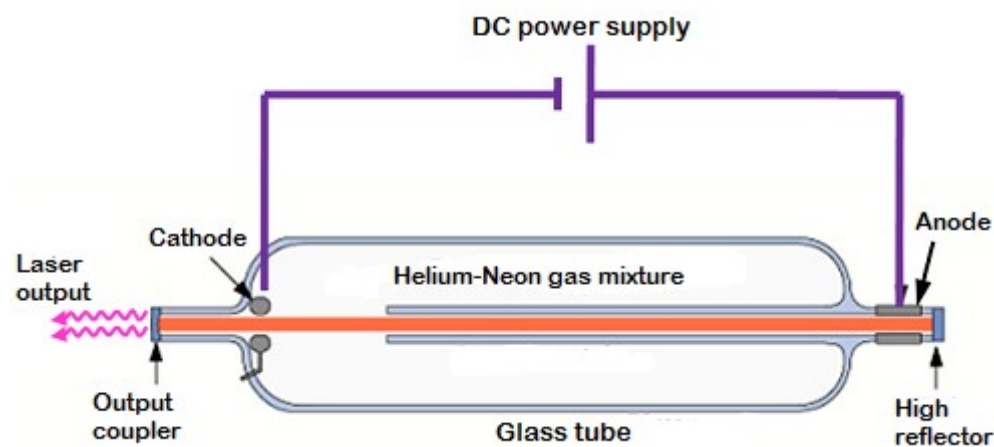
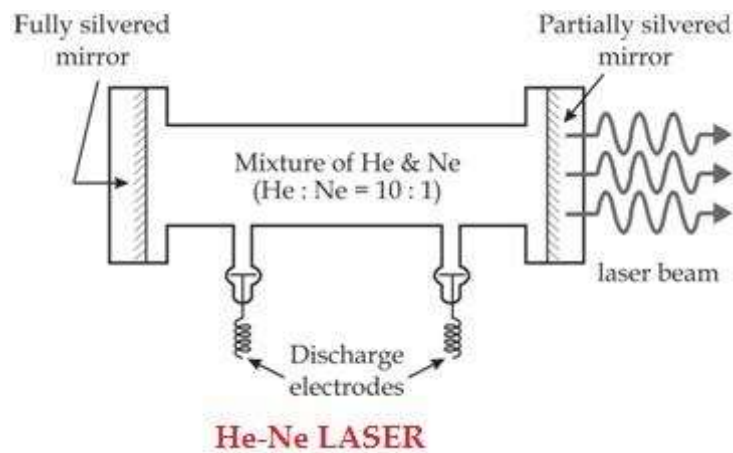


Helium -Neon (He-Ne) Laser

Construction:

(i) Active medium:

It is a gas laser, which consists of a narrow quartz tube filled with a mixture of helium and neon gases in the ratio 10:1 respectively, at low pressure (~ 0.1 mm of Hg). Ne atoms act as active centres and responsible for the laser action, while He atoms are used to help in the excitation process. The length of the quartz tube is about 50 cm and the diameter is about 1 cm.



ii) Optical resonator:

To construct the optical resonator cavity, two parallel mirrors are placed at the ends of the quartz tube one of them is partly transparent while the other is fully reflecting. The spacing between the mirrors is adjusted such that it should be equal to the integral multiple of half-wavelengths of the laser light.

(iii) Pumping system:

The pumping is done through electrical discharge by using electrodes that are connected to a high frequency alternating current source.

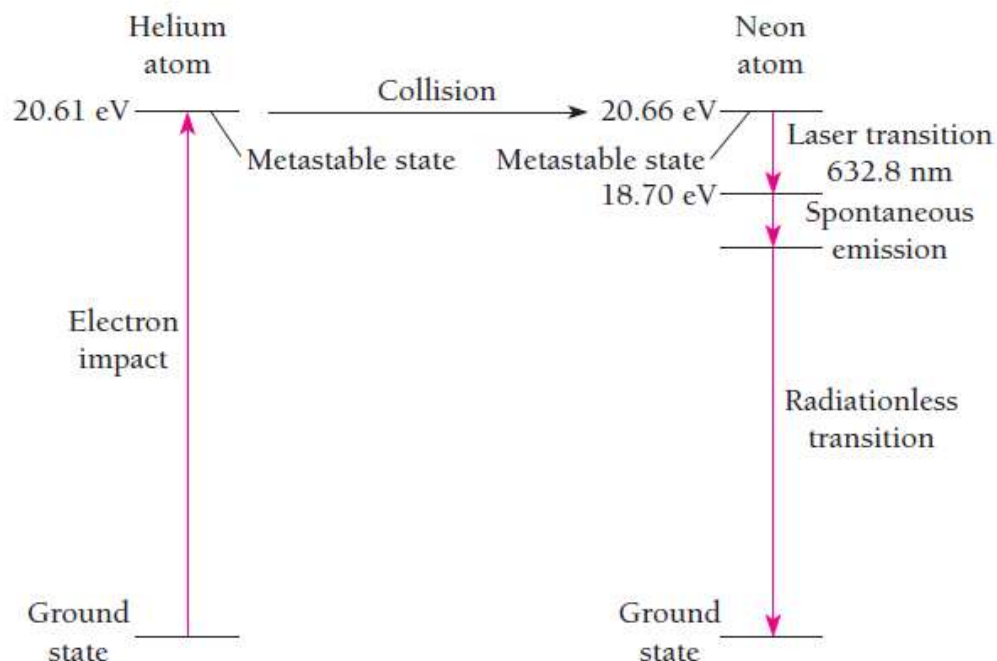
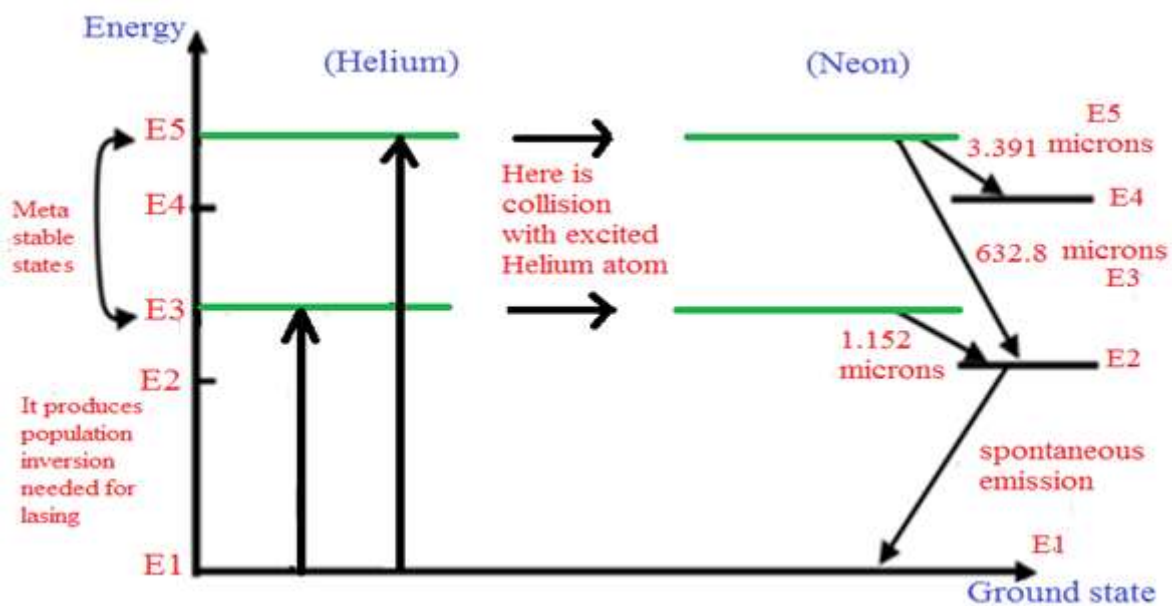


Figure: The helium-neon laser. In a four-level laser such as this, continuous operation is possible. Helium-neon lasers are commonly used to read bar codes.

The common helium-neon gas laser achieves a population inversion in a different way. A mixture of about 10 parts of helium and 1 part of neon at a low pressure (1 torr) is placed in a glass tube that has parallel mirrors, one of them partly transparent, at both ends. The spacing of the mirrors is again (as in all lasers) equal to an integral number of half-wavelengths of the laser light. An electric discharge is produced in the gas by means of electrodes outside the tube

connected to a source of high-frequency alternating current, and collisions with electrons from the discharge excite He and Ne atoms to metastable states respectively 20.61 and 20.66 eV above their ground states. Some of the excited He atoms transfer their energy to ground-state Ne atoms in collisions, with the 0.05 eV of additional energy being provided by the kinetic energy of the atoms. The purpose of the He atoms is thus to help achieve a population inversion in the Ne atoms.

The laser transition in Ne is from the metastable state at 20.66 eV to an excited state at 18.70 eV, with the emission of a 632.8-nm photon. Then another photon is spontaneously emitted in a transition to a lower metastable state; this transition yields only incoherent light. The remaining excitation energy is lost in collisions with the tube walls. Because the electron impacts that excite the He and Ne atoms occur all the time, unlike the pulsed excitation from the xenon flash lamp in a ruby laser, a He-Ne laser operates continuously. This is the laser whose narrow red beam is used in supermarkets to read bar codes. In a He-Ne laser, only a tiny fraction (one in millions) of the atoms present participates in the laser process at any moment.

Characteristics of He-Ne Laser

The He-Ne laser is a relatively low power device with an output in the visible red portion of the spectrum. The most common wavelength produced by He-Ne lasers is 632.8nm, although two lower power (1.152μm and 3.391μm) infrared wavelengths can be produced if desired. Majority of He-Ne lasers generate less than 10m watt of power, but some can be obtained commercially with up to 50m watts of power. For He-Ne lasers the typical laser tube is from 10 to 100 cm in length and the life time of such a tube can be as high as 20,000 hours.

Applications / Uses of He-Ne Laser

The Helium-Neon gas laser is one of the most commonly used laser today because of the following applications.

- ❖ He-Ne lasers are produced in large quantities from many years.
- ❖ Many schools / colleges / universities use this type of laser in their science programs and experiments.
- ❖ He-Ne lasers also used in super market checkout counters to read bar codes and QR codes.
- ❖ The He-Ne lasers also used by newspapers for reproducing transmitted photographs.
- ❖ He-Ne lasers can be use as an alignment tool.
- ❖ It is also used in Guns for targeting.

Advantages of He-Ne Laser

- ❖ He-Ne laser has very good coherence property.
- ❖ He-Ne laser can produce three wavelengths that are 1.152 μ m, 3.391 μ m and 632.8nm, in which
- ❖ the 632.8nm is most common because it is visible usually in red color.
- ❖ He-Ne laser tube has very small length approximately from 10 to 100cm and best life time of 20.000 hours.
- ❖ Cost of He-Ne laser is less from most of other lasers.
- ❖ Construction of He-Ne laser is also not very complex.
- ❖ He-Ne laser provide inherent safety due to low power output.

Disadvantages of He-Ne Laser

The weak points of He-Ne laser are

- ❖ It is relatively low power device means its output power is low.
- ❖ He-Ne laser is low gain system/ device.
- ❖ To obtain single wavelength laser light, the other two wavelengths of laser need suppression, which is done by many techniques and devices. So it requires extra technical skill and increases the cost also.
- ❖ High voltage requirement can be considered its disadvantage.
- ❖ Escaping of gas from laser plasma tube is also its disadvantage.

Semiconductor Diode Laser

Definition:

It is specifically fabricated p-n junction diode. This diode emits laser light when it is forward biased.

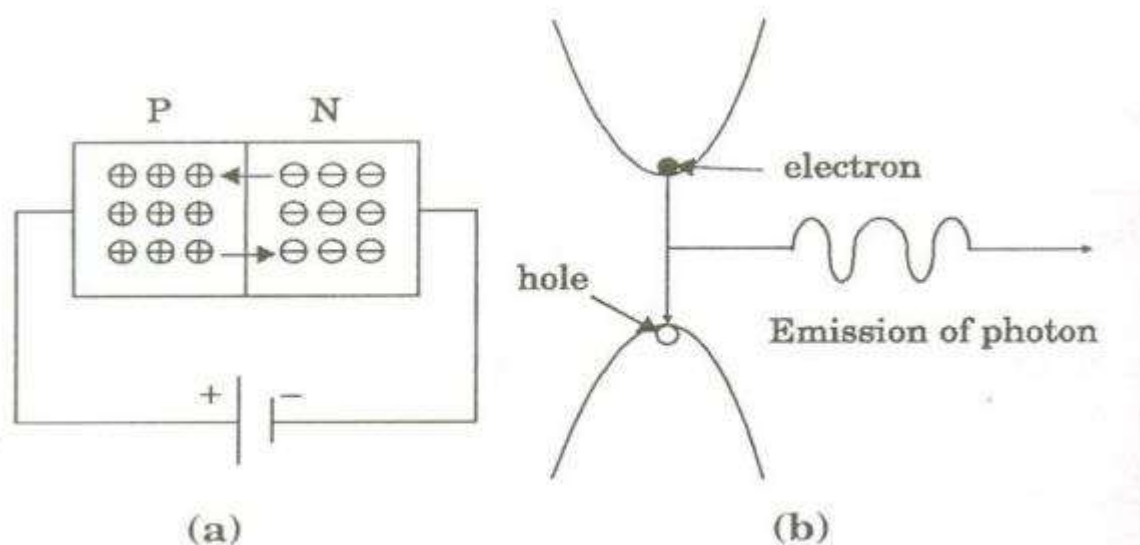
Characteristics:

- ❖ Type: It is a solid state semiconductor laser.
- ❖ Active medium: A PN junction diode made from single crystal of gallium arsenide is used as an active medium.
- ❖ Pumping method: The direct conversion method is used for pumping action
- ❖ Power output: The power output from this laser is 1mW.
- ❖ Nature of output: The nature of output is continuous wave or pulsed output.
- ❖ Wavelength of Output: gallium arsenide laser gives infrared radiation in the wavelength 8300 to 8500 A

Principle:

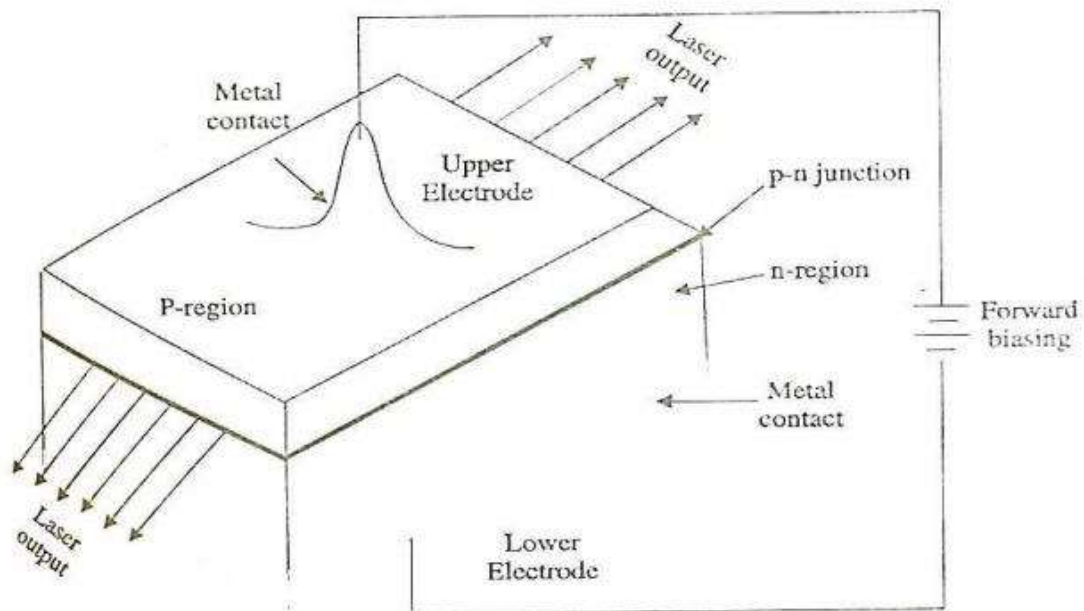
When a p-n junction diode is forward biased, the electrons from n – region and the holes from the p- region cross the junction and recombine with each other.

During the recombination process, the light radiation (photons) is released from a certain specified direct band gap semiconductor like Ga-As. This light radiation is known as recombination radiation. The photon emitted during recombination stimulates other electrons and holes to recombine. As a result, stimulated emission takes place which produces laser.



Construction:

Figure shows the basic construction of semiconductor laser. The active medium is a p-n junction diode made from the single crystal of gallium arsenide. This crystal is cut in the form of a platter having thickness of $0.5\mu\text{mm}$.



The platelet consists of two parts having an electron conductivity (n-type) and hole conductivity (p-type). The photon emission is stimulated in a very thin layer of PN junction (in order of few microns). The electrical voltage is applied to the crystal through the electrode fixed on the upper surface. The end faces of the junction diode are well polished and parallel to each other. They act as an optical resonator through which the emitted light comes out.

Working:

Figure shows the energy level diagram of semiconductor laser.